

Jack Quirion

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EDUCATION

University of Waterloo

Waterloo, ON, CAN

Master of Mathematics in Combinatorics and Optimization

Jan. 2026 – Present

- Specialization: Graph theory
- Supervisor: Prof. Penny Haxell

University of Ottawa

Ottawa, ON, CAN

Joint Honours Bachelor of Science in Computer Science and Mathematics (CO-OP)

Sept. 2021 – Dec. 2025

- CGPA: 9.95/10
- Awarded the Silver Medal for second highest GPA in the Faculty of Engineering
- *The Directed Oberwolfach Problem: Small Cases — A Computational Approach* (Prof. Mateja Šajna)
- *Connected Dominating Sets in Triangulations* (Prof. Vida Dujmović and Prof. Pat Morin)

RESEARCH EXPERIENCE

Game Theory Research Intern

May 2025 – Aug. 2025

Institute of Science and Technology Austria (ISTA)

Klosterneuburg, Lower Austria, AUT

- Member of the Chatterjee Group under the supervision of Dr. Krishnendu Chatterjee.
- Studying the complexity of multi-objective reachability in turn-based stochastic games.

Operations Research Intern

Jan. 2024 – May 2024

Defence Research and Development Canada

Ottawa, ON, CAN

- Developed a differential equations model to simulate training, mentor-mentee and attrition dynamics within fleets of the Royal Canadian Air Force (RCAF).
- Created and deployed a tool in Python based on this model for use by decision-makers in the RCAF, enabling custom scenario analysis with adjustable parameters and showcasing visualizations of key information, such as phase diagrams of personnel states through time and areas of system sustainability.

PUBLICATIONS

A mentored experience accumulation differential model: rapid parameter space analysis applied to Royal Canadian Air Force pilot production, absorption, and retention

Winter Simulation Conference (WSC), 2024, <https://doi.org/10.1109/WSC63780.2024.10838854>

With S. Okazawa, R.M. Bryce, and J.A. Henderson

TEACHING

Teaching Assistant

University of Waterloo

- MATH 128 – Calculus 2 for the Sciences Spring 2026
- Mathematics Tutoring Centre Spring 2026
- MATH 136 – Linear Algebra I for Honours Mathematics Winter 2026

Teaching Assistant

University of Ottawa

- ITI 1120 – Introduction to Computing I Fall 2025
- MAT 1720 – Calcul différentiel et intégral I (Calculus 1) Fall 2025

AWARDS

- **Student Mobility Scholarship**, University of Ottawa 2025
- **Power Corporation of Canada International Experience Scholarship**, University of Ottawa 2025
- **Dean's Honour List**, University of Ottawa 2021-2025
- **CO-OP Student of the Year**, University of Ottawa - Faculty of Engineering 2024
- **CO-OP Student of the Year (Second place)**, University of Ottawa 2024

OTHER EXPERIENCE

Tutor

Aug. 2022 – Present

Scholars Education

Ottawa/Waterloo, ON, CAN

- Provided individualized tutoring in mathematics, computer science and physics to high school students, adapting lesson plans to diverse learning styles.
- Monitored student progress and collaborated with parents and staff to create strategies that improved student confidence and academic performance.

AI/ML Engineering Intern

Sept. 2024 – Dec. 2024

Raven Connected Inc.

Ottawa, ON, CAN

- Developed and implemented event detection algorithms (e.g. refueling) for use on vehicle video telematics devices.
- Built a real-time snap-to-road feature with intersection, crosswalk and speed camera detection, including data preprocessing, cloud communication with AWS and model implementation on Android device.

Data Science Intern

May 2023 – Aug. 2023

Communications Research Centre Canada

Ottawa, ON, CAN

- Designed, implemented, and trained a sophisticated Multi-Agent Reinforcement Learning (MARL) model utilizing PettingZoo and RLlib to simulate bid generation strategies in a dynamic spectrum auction environment.
- Conducted comprehensive analyses of the auction format and procedures leveraging the MARL model outputs, identifying key areas for enhancement and proposing actionable recommendations to improve auction effectiveness.

TECHNICAL SKILLS

Languages: Python, Java, SQL, MATLAB, Go, C++, C, LaTeX, HTML, CSS

Developer Tools: Git, GitHub, Anaconda, Jupyter, AWS, Android Studio, Linux

Libraries: pandas, NumPy, Matplotlib, RLlib, TensorFlow, PyTorch, scikit-learn, SciPy

LANGUAGES

English: Native, **French:** Fluent